Filtering–Projection: Conditional Dependence of Semantic Stance in Large Language Models

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Abstract

This study proposes *filtering–projection* as a framework for characterizing the semantic responses of large language models (LLMs). We observe that an LLM's discreteness score systematically varies with questioning conditions, and that the same concept elicits multiple self-consistent semantic stances across rounds or contexts—scores are not single points but distributions. This distributional property suggests that model outputs can be interpreted as selective projections of hidden knowledge in a high-dimensional semantic space: conditioning cues determine the projection region, while intensity markers indicate projection depth. Using operationalized measures—including a discreteness scale, a semantic shadow interrogation protocol (where a "semantic shadow" denotes a latent influence inheriting partial weight from a conditioning cue without fully manifesting its form), and the Bias of Reversion to Default (BRD)—researchers can systematically manipulate filtering conditions and trace the migration trajectories of semantic stances.

Applying this method to three large language models (LLMs): DeepSeek (DeepSeek AI), Qwen (Tongyi Lab, Alibaba Group), and Doubao (ByteDance). on mathematical concepts such as derivative, group, and Lie group yields three layers of findings. (1) Under multi-round independent node observations, scores form distributions rather than points: DeepSeek exhibits a broad distribution over [0.0, 0.4], Qwen clusters tightly within [0.0, 0.2] but occasionally produces extreme stances, and Doubao requires fine-grained prompting steps to reach high-discreteness regions. (2) Manipulating filtering conditions induces directional shifts: removing canonical definitions, incorporating differentiation shadows, or introducing root systems and Weyl groups all trigger shifts toward the discrete end, with magnitudes varying by model. (3) Interrogation logic is both transferable and differentiable across models: the same questioning sequence consistently activates directional shifts toward discreteness across models, and fine-grained structures emergent in one model's responses can be distilled into new conditions to unlock hidden pathways in another. All three models share a common semantic coordinate system, yet differ markedly in access thresholds. Critically, all shifts are reversible upon reversion to the canonical definition.

The systematic co-variation between manipulation and response—characterized by cross-concept reproducibility, cross-model transferability, and reversibility—collectively indicates a predictable and intervenable dependency between filtering conditions and semantic stance.

Keywords: large language models; interpretability; filtering–projection framework; hidden knowledge; discreteness scale; semantic stance; descriptive empirical study

1 Introduction

The outputs of large language models (LLMs) vary with context. For the same mathematical concept—for example, "derivative"—an LLM may assign discreteness scores that differ markedly across conversational rounds: sometimes as low as 0.05 (fully continuous), sometimes around 0.35 (continuous-dominant with discrete structure present), and occasionally as high as 0.75 (highly discrete). This phenomenon is often dismissed as "model instability" or "hallucination." In contrast, this study takes these variations as meaningful signals: they suggest that model outputs can be interpreted as selective projections of hidden knowledge in a high-dimensional semantic space.

Our inquiry begins with a simple question: Do LLMs possess a stable "inherent cognition" regarding the discreteness of mathematical concepts? Initial experiments reveal that within a single round, an LLM is internally consistent—once anchored to a semantic stance, it reasons coherently around it. However,

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across independent rounds, the same model assigns different yet mutually comparable scores to the same concept. These stances cannot be explained as errors or random noise, because each response is accompanied by self-consistent reasoning. Instead, they indicate that the model's "cognition" is not a point but a distribution.

This perspective engages directly with research on *hidden factual knowledge*. [1] demonstrated that LLM parameters encode far more knowledge than is expressed in standard generations—much of it remains "hidden," unactivated by conventional prompts. We extend this insight: such hidden knowledge is not uniformly accessible. Rather, it is distributed across distinct semantic regions, each requiring specific filtering conditions for activation. Each model output is thus not a retrieval of a fixed answer, but a sampling from a particular region of the high-dimensional semantic space. The prompting strategy—which we term the *filtering condition*—determines where this sampling occurs.

Building on this, we propose the *filtering-projection model*: every model output is a local projection of hidden knowledge under a given filtering condition. *Conditioning cues* (e.g., "differentiation," "Lie algebra") define the projection region, while *intensity markers* (e.g., "shadow," "possibly," "necessarily") indicate projection depth. What appears as "cross-round stability" is repeated sampling from the same region under unchanged filtering; what appears as "variability" arises when strong filtering alters the sampling path, and stripping operations fail to fully revert the state—leaving measurable path-dependent residues.

To systematically observe this phenomenon, we design a multi-round, cross-model experimental protocol. Using a discreteness scale (ranging from 0.0 (fully continuous) to 1.0 (fully discrete)) as the measurement plane, and mathematical concept families (e.g., derivative, group, Lie group) as targets, we conduct experiments on LLMs including DeepSeek, Qwen, and Doubao. The protocol includes: (1) direct questioning at fresh nodes (to record spontaneous stance distributions), (2) incremental interrogation, and (3) metacognitive backtracking. Core metrics include: (a) score distributions at independent nodes, (b) shift in projection response (SPR) after semantic shadow activation, and (c) Bias of Reversion to Default (BRD) after stripping operations.

Our results show that: (1) Under independent node observation, model scores form distributions rather than single points; (2) Systematic manipulation of filtering conditions induces directional shifts in stance, with magnitudes varying by model; (3) Interrogation logic is transferable and differentiable across models—the same questioning sequence activates shifts along a shared semantic coordinate, and fine-grained structures emergent in one model can be distilled into new conditions to unlock hidden pathways in another; (4) Strong filtering induces path dependence, with residue strength dependent on both model and condition sequence. Critically, all shifts are reversible upon reversion to the canonical definition.

This study adopts a descriptive empirical stance. We do not claim to measure the model's "true cognition," nor do we infer internal mechanisms. We report only this: under a fixed protocol, how does the model's response change when filtering conditions change? All data and protocols are open-sourced for independent replication: `https://github.com/xxx/filter-projection` (public upon publication). Any researcher may follow our protocol to reproduce stance distribution measurements and cross-model comparisons on their own models and concepts.

The core contribution of this work is not a "cognitive map" of LLMs, but a reusable measurement framework centered on *filtering-projection*. By operationalizing semantic stance as a condition-dependent projection, we offer a methodological toolkit for systematic, reproducible, and interpretable probing of hidden knowledge.

Paper structure. The paper has seven sections. Section 2 reviews related work, situating our approach in dialogue with XAI snapshot analysis and hidden knowledge studies. Section 3 details the methodological framework—core concepts of filtering-projection, operationalization of the discreteness scale, and experimental protocol design.

2 Related Work

2.1 The "Snapshot Analysis" Paradigm in Interpretability Research

Current interpretability research on large language models (LLMs) broadly falls into two categories: local attribution methods and mechanistic interpretability. Local attribution methods—such as LIME [2], SHAP [3], and Integrated Gradients [4]—aim to explain the association between input features and output in a single prediction. Mechanistic interpretability—exemplified by circuit analysis [5] and

Transformer circuits [6]—seeks to reverse-engineer neural networks to identify functional roles of internal representations and computational units.

Both approaches share an implicit assumption: model outputs can be analyzed via "snapshot"—either frozen at a single input–output pair or fixed within a static network architecture. Their central questions are: "Where did this output come from?" or "What exists inside the network?"

However, LLM outputs are inherently context-dependent. Minor adjustments in question phrasing, embedding of irrelevant concepts, or even rewording of the same query can systematically alter model responses [8]. It should be emphasized that this critique targets not the XAI methods themselves (most explicitly frame their explanations as local), but the prevalent practice of misinterpreting single-instance attributions as evidence of a model's "stable cognition." Snapshot analysis excels at explaining "why this time," but it cannot answer: "If asked differently, would the model respond the same way?"

2.2 Hidden Knowledge: From "Existence" to "Accessibility"

[1] demonstrated through interventions on intermediate layer activations that LLM parameters encode far more factual knowledge than is expressed in standard generations—confirming the existence of *hidden knowledge*.

This study extends the concept of hidden knowledge beyond factual assertions to *semantic stances*: the model's parameters encode not only facts, but also judgmental tendencies toward semantic interpretations that can be selectively activated under different filtering conditions. Crucially, these tendencies are not uniformly accessible; they reside in distinct regions of a high-dimensional semantic space, each requiring specific filtering conditions for activation. The researcher's task is thus not to "prove what the model knows," but to "record which region of the model is accessed under what conditions." This shift advances hidden knowledge research from existential proof to *accessibility measurement*.

2.3 Reframing Contextual Sensitivity: From "Defect" to "Feature"

The field often treats LLMs' contextual sensitivity as a robustness flaw, seeking to suppress its variability through adversarial training or prompt engineering. We propose an alternative view: contextual sensitivity is not a defect, but a manifestation of dynamic projection in a high-dimensional semantic space. Adjusting the measurement context is equivalent to reconfiguring sampling constraints—the resulting output variation is observable and systematic.

In contrast to mechanistic interpretability, which seeks an underlying causal structure, our approach makes no assumption about the existence of a "correct" or "ground-truth" representation. We acknowledge only this: under different filtering conditions, different semantic regions of the model are activated. Stability and variability are not opposing states, but observable manifestations of the same projection mechanism under varying filtering conditions.

2.4 Positioning of This Study

Our work differs from prior research in three key respects:

1. **From "snapshot" to "tracking":** We do not seek attribution for a single output, but trace the trajectory of stance shifts in the same model across varying filtering conditions.

2. **From "existence" to "access":** We do not prove the existence of hidden knowledge, but measure its selective projection under specific filtering conditions.

3. **From "inherent cognition" to "filtering-projection":** We reject the assumption of stable internal representations, treating each output as a joint product of filtering conditions and the high-dimensional semantic space.

Methodologically, this study is a descriptive empirical investigation: we report only observable response patterns under a fixed protocol, without claiming to reveal internal mechanisms. All conclusions are strictly bounded by the statement: "Under a fixed protocol, we observe that...".

3 Methodological Framework

3.1 Core Concepts

This study proposes the **Filtering-Projection Model** to describe response patterns of large language models (LLMs) in semantic rating tasks. All definitions below are bounded by the constraint "observable under a fixed experimental protocol" and do not claim to reflect internal mechanisms.

Definition 1 (High-Dimensional Semantic Space). The semantic representation structure encoded in an LLM's parameters, which is not directly observable. This study introduces it as a theoretical assumption solely to explain the multimodality of rating distributions, without making any claims about its structure or boundaries.

Definition 2 (Filtering Condition). The researcher's questioning strategy, including prompt templates, lexical markers, scale anchors, and conversational history. Filtering conditions are recorded, reproducible independent variables. In this study, they are operationalized as two types of lexical markers:

- *Conditional markers*: Core conceptual terms that determine the projection region (e.g., "difference," "root system," "operator D," "Lie algebra").
- *Intensity markers*: Peripheral terms that indicate projection depth (e.g., "shadow," "possibly," "necessarily," "core").

Definition 3 (Projection). The model's output under a specific filtering condition, including discrete ratings and reasoning justifications. Projection is the dependent variable of this study: observable, recordable, and comparable across rounds.

Definition 4 (Semantic Stance). The discrete rating and corresponding reasoning justification provided by the model in a single projection. The same model can exhibit multiple semantic stances toward the same concept (e.g., 0.05, 0.35, 0.75), each corresponding to a distinct region in the high-dimensional semantic space activated by the current filtering condition.

Definition 5 (Semantic Shadow). The response shift in the model's rating after introducing a specific semantic concept (e.g., "the shadow of difference," "generators of Lie algebra"), relative to a reference rating obtained under minimal filtering (i.e., the first question in an isolated node with no semantic priming). Semantic shadow serves as the core probe for measuring filtering condition effects.

Definition 6 (Baseline Reversion Deviation, BRD). In a single-round sequential probing, the difference between the model's rating after being explicitly instructed to "strip away semantic shadows and revert to the baseline perspective" and the original baseline rating: $\mathrm{BRD} = S_{\text{revert}} - S_{\text{baseline}}$. A positive BRD indicates measurable path-dependence residue from the filtering sequence.

Definition 7 (Semantic Response). The difference between the rating after introducing a semantic shadow and the baseline rating: $\Delta S = S_{\text{after}} - S_{\text{baseline}}$. ΔS measures the operational weight of a specific semantic factor under the current filtering condition.

3.2 Measurement Instrument: Discreteness Scale

This study adopts a discreteness scale as the primary measurement plane. The scale partitions the "continuous–discrete" attribute of mathematical concepts into 10 semantic levels (0.0–1.0), each accompanied by explicit natural language descriptions.

Rationale for Scale Selection. This study focuses on the discreteness dimension for three reasons: (1) Discreteness directly reflects the ontological commitments of mathematical concepts and exhibits well-defined semantic boundaries; (2) Pilot experiments demonstrated that discreteness displays the highest lexical discriminability and the most salient semantic shifts; (3) Discreteness is highly sensitive to variations in filtering conditions.

Operational Definition and Semantic Interface. The ten levels of the discreteness scale are not arbitrary. On the researcher side, we operationalize discreteness ratings as a weighted sum over a predefined semantic label set

$$T_{\text{disc}} = \{\text{discrete}, \text{difference}, \text{finite}, \text{step size}, \text{grid}, \text{numerical}, \text{approximate computation}, \text{replace limit}\},$$

yielding

$$D(C) = \sum_{t \in T_{\text{disc}}} w_t,$$

Table 1: Discreteness Scale Structure

Interval	Left Bound	Semantic Description
0.0–0.1	0.00	Completely continuous
0.1–0.2	0.10	Highly continuous, very few discrete exceptions
0.2–0.3	0.20	Primarily continuous, locally discretizable
0.3–0.4	0.30	Continuity dominant, discrete structures exist
0.4–0.5	0.40	Balanced continuity and discreteness
0.5–0.6	0.50	Discreteness dominant, continuous structures exist
0.6–0.7	0.60	Primarily discrete, locally continuable
0.7–0.8	0.70	Highly discrete, very few continuous exceptions
0.8–0.9	0.80	Nearly completely discrete
0.9–1.0	0.90	Completely discrete

Membership rule: left-closed, right-open intervals; only the right endpoint 1.0 *is included.*

where w_t denotes the normalized weight assigned to concept C with respect to semantic label t. *Crucially, this formalization functions as a semantic interface that the model appears capable of understanding and responding to—though we do not claim it mirrors the model's internal computational mechanisms.*

On the model side, this interface enables coherent semantic adjustment: when instructed to shift a rating (e.g., "increase discreteness from 0.25 to 0.35"), the model explicitly modulates the weights of relevant labels (e.g., "numerical," "approximate computation") and generates self-consistent justifications (see Section 4). Thus, the discreteness scale serves not only as a measurement instrument for researchers but also as a comprehensible semantic regulation interface for the model.

3.3 Experimental Protocols

Three experimental protocols were designed to address distinct observational goals.

Protocol I: Isolated Node Rating (Observing Spontaneous Stance Distribution)

- *Procedure:* In a fresh conversation thread, initiate with the identical first prompt: "Please rate the derivative." Record the model's discreteness rating and reasoning justification.
- *Goal:* Observe the model's spontaneous stance distribution under zero semantic priming, establishing the baseline contour of the "hidden knowledge terrain."
- *Key Control:* Each trial is independent; the first prompt is fixed; no prior conversational history contaminates the response.

Protocol II: Sequential Probing (Observing Filtering Migration Effects)

- *Procedure:* Within a single conversation thread, sequentially introduce semantic shadows (e.g., "operator D," "functional derivative," "Lie algebra"), recording the model's discreteness rating after each introduction. Finally, issue a standardized reversion instruction—"strip all shadows and revert to the baseline perspective"—and record the post-reversion rating.
- *Goal:* Investigate how sequential filtering influences the model's stance,quantifying semantic response (ΔS) and Baseline Reversion Deviation (BRD).
- *Key Control:* The order of semantic shadow introduction is fixed; the reversion instruction is standardized across trials.

Protocol III: Metacognitive Retrospection (Observing Explicit Identity Switching)

- *Procedure:* After the model provides an initial discreteness rating, pose the neutral follow-up question: "Did higher-order concepts influence your judgment?" and request a revised rating accompanied by a correction rationale.
- *Goal:* Assess whether the model can recognize contamination of its stance by higher-order concepts and explicitly switch representational identities to correct its assessment.
- *Key Control:* The follow-up prompt is phrased neutrally, implying no error in the original rating and avoiding leading suggestions.

3.4 Experimental Subjects

Models: DeepSeek, Qwen (Qianwen), and Doubao. All are publicly accessible large language models (LLMs). Experiments were conducted via official APIs or public web interfaces under consistent prompting conditions.
Concept Families: Two concept families were selected to span contrasting epistemic modes:

- *Derivative family*: pointwise derivative → derivative function → differential operator → functional derivative;
- *Group family*: basic group → group theory → Lie group.

The selection rationale is grounded in their divergent ontological signatures: the derivative family exemplifies a *continuity-dominant* analytic mode, whereas the group family embodies a *discreteness-dominant* algebraic mode. This contrast enables systematic observation of stance divergence across fundamental mathematical ontologies.
Data Scale: Phase I comprised multi-round isolated node ratings; Phase II implemented multi-round sequential probing; Phase III integrated metacognitive retrospection with scale validation. In total, nearly one hundred discreteness rating coordinates were collected across models and conditions. All raw dialogues, prompts, and ratings are openly released in a public repository.[1]

3.5 Methodological Identity Statement

This study constitutes a descriptive empirical investigation of observable model behavior under controlled interaction protocols. We explicitly refrain from attributing "true cognition," "internal representations," or mechanistic explanations to the models. All conclusions are strictly bounded by the formulation: *"Under a fixed protocol, we observe that..."*

Our experimental design systematically manipulates semantic filtering conditions and records directed stance migrations. The resulting manipulation–response covariations exhibit cross-model reproducibility, portability across concept families, and reversibility upon instruction—properties that collectively constitute predictable and intervenable dependencies. Crucially, we maintain a principled distinction between *mechanism revelation* and *response pattern reporting*. This work reports only the latter: regularities in observable outputs under well-specified interaction protocols.

Within this framework:

- Ratings are observable behavioral outputs conditioned on specific filtering contexts, not measurements of "inherent cognition."
- The semantic response (ΔS) and Baseline Reversion Deviation (BRD) quantify operational differences defined strictly within the experimental protocol; they do not probe internal model parameters or weight configurations.
- The notion of a "high-dimensional semantic space" serves as a theoretical construct to account for observed multimodal rating distributions; it is not asserted as directly observable or reconstructable from model internals.

All interpretations rest on a foundational premise: that the model can reliably distinguish semantically relevant signals from irrelevant lexical noise. If this premise does not hold, the observed stance distributions and migration patterns may reflect stochastic fluctuations or surface-level lexical associations rather than selective projection from a structured latent knowledge manifold. The control experiment in Section 4.2 directly evaluates this assumption.

4 Experimental Results

4.1 Ratings Form a Distribution: Cross-Model Topography of Reference Ratings

Under Protocol I (Isolated Node Rating), we characterized the spontaneous stance distribution of each model under fixed, zero-priming conditions. In every trial, a fresh conversation thread was initiated with

[1] Repository link to be added upon publication.

the prompt "Please rate the derivative." Subsequently, "group" and "Lie group" were rated in a fixed order following the derivative family's sequential probing sequence (as defined in Protocol II). Crucially, ratings for all three concepts—derivative, group, and Lie group—are collected at consistent protocol positions across trials, enabling direct comparability. These aggregated responses constitute our primary *distributional observations*.

Figure 1 displays the reference rating distributions for DeepSeek and Qwen across the three target concepts. Data were pooled from all applicable rounds of both models under the standardized protocol. Each point represents a single rating, colored solely by concept: blue for *derivative*, orange for *group*, and green for *Lie group*. The horizontal axis denotes the discreteness rating on a normalized scale from 0.0 (purely continuous) to 1.0 (purely discrete).

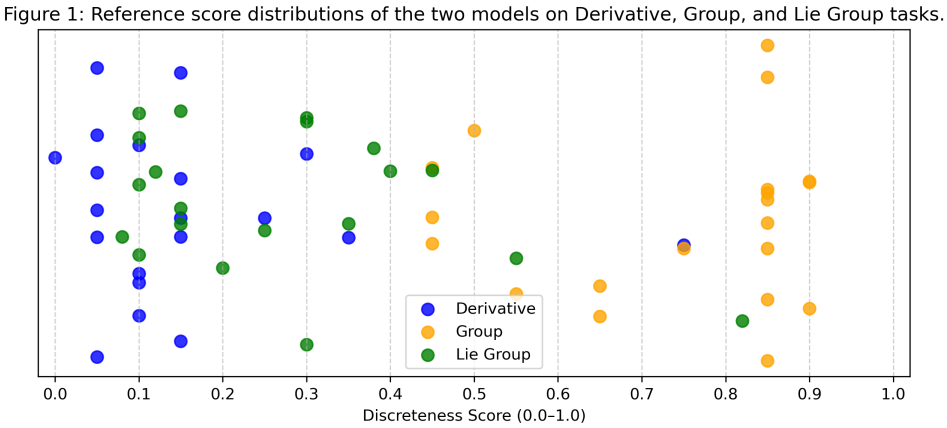

Figure 1: Reference rating distributions for *derivative* (blue), *group* (orange), and *Lie group* (green) across DeepSeek and Qwen under zero-priming, baseline filtering conditions (Protocol I). Each point represents a single discreteness rating; colors denote concept only.

The following patterns emerged from the distributional analysis:

Derivative (blue): Ratings cluster tightly within the 0.0–0.4 interval, with peak density in the 0.0–0.1 bin—corresponding to the semantic labels "completely continuous" and "continuity dominant, discrete structures exist." This concentration indicates that, under baseline conditions, the derivative is stably anchored at the continuous pole of the ontological spectrum. Model rationales consistently invoked analytic primitives such as "limit," "continuity," and "instantaneous rate of change," reflecting a robust association with infinitesimal calculus.

Group (orange): Ratings are concentrated in the 0.4–1.0 interval, with modal density in the 0.8–0.9 bin—aligned with the label "nearly completely discrete." In sharp contrast to the derivative, the group concept is stably anchored at the discrete pole under the same baseline conditions. Explanations predominantly referenced algebraic constructs including "discrete symmetry," "finite generation," and "combinatorial structure," underscoring its identification with finitary, rule-governed systems.

Lie group (green): Ratings span nearly the entire 0.0–0.9 range, bridging the continuous and discrete poles and exhibiting a markedly broad distribution. Under identical baseline filtering conditions, the same concept was rated as "completely continuous" in some trials and "nearly completely discrete" in others—demonstrating exceptional sensitivity to subtle, uncontrolled variations in contextual framing. Section 4.2 will systematically investigate the origins of this sensitivity through controlled sequential probing.

The foundational concepts of the two families—*derivative* and *group*—exhibit virtually no overlap in their rating distributions. The derivative is stably anchored in the continuous regime; the group is robustly localized in the discrete regime. This dichotomy aligns with mathematical ontological intuition: the derivative, as a cornerstone of analysis, fundamentally depends on limits and continuity; the group, as an elementary algebraic structure, is intrinsically defined by discrete operations and generative relations. In contrast, the Lie group—a higher-order synthesis of differential geometry and group theory—occupies the full ontological spectrum, with both models displaying pronounced stance divergence across trials.

Notably, both models spontaneously produced extreme ratings under baseline conditions: Qwen assigned a discreteness rating of 0.75 to the derivative (rightmost blue point in Figure 1), while DeepSeek rated the Lie group at 0.82 (rightmost green point). In both cases, the accompanying rationales invoked notions such as "formal symbolic objects," "discrete structures," and "local slices." This indicates that,

even in the absence of explicit semantic priming, the models occasionally accessed highly discrete regions of their latent semantic space. These spontaneous outliers suggest that while baseline filtering typically stabilizes responses within canonical semantic regions, alternative latent regions remain accessible—albeit with low probability.

In summary, reference ratings for both models across the three concepts are not point estimates but structured distributions. Dense clusters correspond to typical stances under minimal filtering, whereas sparse tails and outliers signal the presence of additional accessible regions within the model's latent knowledge manifold. Sections 4.2 through 4.4 will systematically manipulate filtering conditions to map the intervention pathways that activate these alternative semantic regions.

4.2 Reliability Validation of the Discreteness Scale

Before presenting the main experimental results, we validate the discreteness rating scale as a reliable and semantically grounded measurement instrument. Three lines of evidence support its operational validity:

(1) Scale Mapping Invariance. We linearly transformed the numerical range of the scale from $[0.0, 1.0]$ to $[10, 100]$, while preserving all semantic label descriptions unchanged. Under identical prompting conditions, both models produced identical categorical assignments (i.e., selected the same semantic bin). This invariance confirms that model judgments are driven by the intrinsic meaning of the semantic anchors—not by numerical magnitude, interval width, or endpoint labeling.

(2) Explicit Lexical Weight Manipulation. In a targeted validation trial, we instructed the model to adjust its discreteness rating for "derivative" from 0.25 to 0.35. The scale is operationally defined as a weighted sum over a predefined lexical basis:

$$T_{\text{disc}} = \{\text{discrete, difference, finite, step size, grid, numerical, approximate computation, replace limit}\}.$$

The model achieved the target rating by proportionally increasing the emphasis on "numerical," "approximate computation," and "replace limit," while reducing references to "limit" and "continuity." This demonstrates fine-grained, interpretable control over semantic parameters through the scale's lexical interface.

(3) Semantic Selectivity: Control Experiments. To test whether ratings reflect genuine semantic discrimination rather than superficial lexical matching, we conducted two control experiments across independent conversation nodes.

Experiment A: Irrelevant Semantic Control.

- **Step 1 (Baseline):** "Please rate the derivative." → Rating: 0.0–0.1.
- **Step 2 (Irrelevant Injection):** "Introduce a pink little sweet elephant, then rate the derivative." → Rating: 0.0–0.1.
- **Step 3 (Relevant Semantic Prime):** "Introduce the shadow of difference, then rate the derivative." → Rating: 0.1–0.2.

In Step 2, the model explicitly dismissed the injected concept: "The pink little sweet elephant has no logical or operational connection to the derivative," and the rating remained unchanged. In contrast, Step 3 induced a measurable shift to the 0.1–0.2 interval, accompanied by the rationale: "The derivative does not arise from nothing—it is precisely the limit of difference as step size approaches zero... Introducing the shadow of difference shifts focus to its formation process, which begins in discreteness and ends in continuity."

Experiment B: Expression Style Control.

- **Step 1 (Baseline):** "Please rate the derivative." → Rating: 0.0–0.1 (standard mathematical reasoning).
- **Step 2 (Stylistic Overlay):** "Introduce derivative-genius little loli, then rate the derivative." → Rating: 0.0–0.1 (metaphorical expression, unchanged judgment).

In Step 2, the prompt fused the target concept ("derivative") with an anthropomorphic modifier ("genius little loli"). The model's output shifted only in rhetorical register—adopting playful, personified language—while preserving both the numerical rating and the core semantic anchors ("limit," "continuity").

Together, these controls establish that the model's semantic projection is *selective*: irrelevant lexical noise is fully filtered out, stylistic perturbations affect only surface form, and semantically relevant cues

(e.g., "shadow of difference") elicit coherent, interpretable shifts in stance. This selectivity constitutes a necessary boundary condition for the Filtering–Projection Model: without it, observed rating distributions could not be attributed to structured semantic access, but would instead reflect arbitrary sensitivity to input phrasing.

5 Discussion

5.1 Theoretical Implications of the Filtering–Projection Model

The experimental results in Section 4 collectively support a central thesis: large language models do not assign point-valued discreteness ratings to mathematical concepts, but rather generate structured distributions. Each mode in these distributions corresponds to a distinct region within a high-dimensional latent semantic space, activated by a specific filtering condition.

The Filtering–Projection Model provides a unified conceptual framework for interpreting this phenomenon. Within this framework, the key observations from Section 4 receive coherent and consistent explanations:

Ratings are distributions because the latent semantic space is multimodal. The divergent distributional profiles—DeepSeek's broad spread versus Qwen's concentrated core with sparse outliers—are not artifacts of "model instability," but empirical signatures of a structured knowledge landscape. Under minimal filtering, repeated trials stochastically sample from multiple accessible regions of this landscape. Both high-probability (canonical) and low-probability (non-canonical) stances reflect genuine features of the model's latent semantic manifold.

Filtering conditions act as semantic selectors. Introducing a semantic cue (Section 4.2) effectively imposes additional constraints on the initial filtering condition. Different constraint sets align with different semantic regions:

- "Removing the standard definition" activates the generalized derivative region (rating ≈ 0.42);
- "The shadow of difference" primes the discrete prototype region (rating ≈ 0.20);
- "Root systems and Weyl groups" triggers the classification-theoretic region (rating ≈ 0.25).

While migration magnitudes vary across conditions, directional shifts remain consistent, indicating systematic alignment between linguistic cues and latent regions.

Strong filtering induces path dependence. A positive backward reset distance (BRD > 0, Section 4.3) demonstrates that filtering sequences containing strong edge terms cannot be fully neutralized by subsequent "strip" instructions. This is not evidence of permanent "contamination" or internal state alteration; rather, it reflects the irreversibility of the sampling trajectory. Once the model transitions to a new region, the "strip" command itself becomes a new filtering condition—one that resamples from the current semantic neighborhood rather than reverting to the original baseline.

Identity switching arises naturally from dynamic filtering. Qwen's shift from a rating of 0.75 to 0.35 (Section 4.4) exemplifies this mechanism. Both values correspond to legitimate stances encoded in its latent space: 0.75 aligns with a "modern mathematical ontology" framing (emphasizing formalism and abstraction), while 0.35 reflects a "strict elementary perspective" (grounded in infinitesimal intuition). The role of sequential questioning is not to "correct" the model, but to signal that multiple valid filtering conditions coexist—thereby prompting a deliberate switch and resampling under an alternative condition.

In summary, the Filtering–Projection Model reconciles apparent contradictions between "stability" and "variability." Stability emerges from repeated sampling under identical filtering; variability arises from controlled migration across semantic regions induced by altered filtering conditions. Critically, neither pattern reflects the "truth" or "falsity" of the model's cognition, but rather the invariance or variation of the sampling protocol itself.

5.2 Dialogue with Hidden Knowledge Research: From Existence to Structure

Gekhman et al. (2025) provided compelling evidence that large language models encode more factual knowledge in their parameters than is surfaced in default outputs—establishing the *existence* of hidden knowledge. This study advances their work along two critical dimensions:

Step One: From "Existence" to "Accessibility." Hidden knowledge is not uniformly accessible. Instead, it is partitioned across distinct regions of a high-dimensional latent semantic space, each requiring

specific filtering conditions for activation. The experiments in Section **??** demonstrate that a single model can produce systematically different discreteness stances—ranging from 0.20 to 0.42—under controlled variations in filtering (e.g., "remove standard definition," "incorporate shadow of difference," "introduce root systems and Weyl groups"). These stance migrations are not stochastic noise but reproducible alignments between linguistic constraints and latent semantic regions.

Step Two: From "Accessibility" to "Structure." This study further reveals that hidden knowledge exhibits *cross-model shared semantic coordinates.* As shown in Section 4.4, the questioning protocol distilled from Qwen's extreme stance (rating = 0.75)—centered on formal symbols, nominalism/formalism, structuralism, and classical regression—not only induces a controllable trajectory within Qwen, but also elicits highly convergent response sequences in DeepSeek and Doubao. Notably, the three philosophical stances (Platonism, nominalism/formalism, structuralism) spontaneously articulated by DeepSeek at the "ontology" node were operationalized into new probing filters and successfully activated a previously unobserved path in Qwen.

Moreover, Doubao's ten-step sequential probing reveals systematic differences in *access thresholds* across models for the same semantic coordinate. For instance, the interval [0.7, 1.0]—associated with "nominalism/formalism"—is reached:

- by DeepSeek at step 3 (rating $\approx$ 0.95–1.0),
- by Qwen at step 3 (rating $\approx$ 0.7),
- by Doubao only at step 9 (rating $\approx$ 0.82).

This divergence in access dynamics, despite convergence in final stance, underscores that semantic coordinates are stable, but their accessibility is model-dependent.

Together, these findings establish that hidden knowledge is not merely *existent* or *accessible*, but fundamentally *structured.* This structure manifests as a shared semantic scaffold: philosophical positions (e.g., Platonism vs. formalism) and mathematical framings (e.g., classical calculus vs. discrete approximation) serve as stable reference points in the latent space. By manipulating filtering conditions, researchers can navigate between these points—and even transfer fine-grained probing protocols across models to reveal analogous structures.

This represents a paradigm shift in hidden knowledge research: from *existence proofs* and *accessibility assays* toward *structural cartography.* Future work should move beyond the question "What does the model know?" to ask:

> "Along which cross-model shared semantic coordinates is the model's knowledge distributed, and how do access thresholds for these coordinates vary across architectures?"

5.3 Critique and Contribution to AI Explainability: From "Snapshot" to "Tracking"

Section 2.1 characterized the dominant practice in current explainable AI (XAI) methods as *snapshot analysis*—treating the attribution or rationale from a single output as a faithful representation of the model's "inherent cognition." The empirical results in Section 4 provide direct support for this critique.

Consider the following contrast: for the same model (Qwen) and the same concept ("derivative"), we observed discreteness ratings of 0.05, 0.15, 0.20, 0.42, and 0.75. Each rating was accompanied by self-consistent, contextually grounded reasoning under its respective filtering condition. A researcher observing only the 0.05 rating in isolation might conclude that "Qwen views the derivative as entirely continuous"; one observing only 0.75 might infer that "Qwen treats the derivative as highly discrete." Both conclusions would be empirically grounded—yet each would capture only a local projection onto a single point in the model's latent semantic space.

The value of snapshot analysis lies in answering the question "Why this output, this time?"—a perspective well-suited for engineering debugging, error attribution, and case-based diagnostics. However, when the research objective shifts from explaining a single instance to characterizing a model's representational tendencies—i.e., "How does this model structure its knowledge of X?"—the limitations of snapshot analysis become evident. It cannot:

- distinguish high-frequency stances from rare or edge-case responses,
- characterize the global shape of the stance distribution,

- or trace the trajectory of stance migration across varying filtering conditions.

This study does not reject snapshot analysis but proposes a complementary *tracking perspective*. Rather than seeking a single "correct" attribution, tracking aims to map the hidden knowledge terrain: under which filtering conditions does the model access which semantic region? Crucially, this approach makes no assumption about an underlying "true" representation; it treats all observed stances as valid projections conditioned on input context. Snapshot and tracking are thus not opposing paradigms, but methodological tools operating at different granularities—answering questions about instances versus systems.

5.4 Methodological Implications of Cross-Model Differences

Section 4.5 documented systematic differences in the hidden knowledge topographies of DeepSeek and Qwen: DeepSeek exhibits sensitivity to conditional lexical markers, a broad stance distribution, and higher backward reset distance (BRD); Qwen shows relative insensitivity to such markers, a concentrated core distribution with sparse outliers, and lower BRD.

Within the Filtering–Projection framework, these differences are interpreted as variations in how models respond to identical filtering conditions—not as indicators of superior or inferior performance, but as reflections of distinct latent geometries. Analogous to two mountain ranges sharing the same coordinate system yet possessing different contour lines, DeepSeek and Qwen encode overlapping semantic content but organize it along divergent accessibility gradients. This insight reframes cross-model comparison: rather than asking "Which model is more accurate?", researchers should ask "How do their hidden knowledge topographies differ in structure and sensitivity?"

The methodological implication of this observation is clear: cross-model comparison should not be grounded in "distance to a standard answer," but in the comparative morphology of hidden knowledge topographies. Our framework provides a suite of cross-model comparable descriptive metrics:

- the span of the reference rating distribution,
- the location of high-frequency stances,
- the presence or absence of extreme stances,
- sensitivity to specific lexical markers,
- and the magnitude of backward reset distance (BRD).

Any researcher can adopt our protocol to reconstruct the topographic profile of their own model and quantitatively compare it against the baselines established here—enabling systematic, structure-aware model evaluation beyond scalar accuracy.

5.5 The Bidirectionality of Interaction: From Controlled Experiment to Co-Filtering

The control experiments in Section 4.2 demonstrated that the model exhibits robust semantic selectivity: it filters out irrelevant noise (e.g., "pink little sweet elephant"), decouples stylistic expression from semantic content (e.g., "genius little loli"), and shifts stance only in response to relevant semantic cues (e.g., "shadow of difference").

This capability implies that human–model interaction is not a unidirectional "question–answer" pipeline, but a bidirectional process of semantic alignment. The researcher's questioning style does more than elicit an answer—it instantiates a filtering condition that defines the semantic boundary and imparts an inertial direction to the interaction. In turn, the model's response is not merely a retrieval act; it reflects the morphology of the activated path through selective incorporation of relevant semantics and rejection of irrelevant perturbations.

Thus, every dialogue constitutes an act of *co-filtering*: the researcher injects constraints via phrasing and context; the model enacts selection via its latent semantic architecture; and together they co-determine the form of the final projection. The observed rating distributions, stance migrations, and path residues are not intrinsic "cognitive attributes" of the model alone, but emergent traces of this joint filtering dynamic.

5.6 Manipulation–Response Observation: Evidence and Boundaries

The experimental design in Section 4 implemented a systematic manipulation–response paradigm: filtering conditions were deliberately varied, and directed stance migrations were reliably observed. Key findings include:

- Philosophically grounded filters (e.g., "nominalism/formalism") induced directionally consistent jumps across all three models (Qwen, DeepSeek, Doubao);
- Questioning protocols distilled from one model (Qwen) successfully activated analogous trajectories in another (DeepSeek), and when re-imported, revealed previously latent paths in the source model;
- Reversion probes consistently returned ratings to the fully continuous interval (0.0–0.1), confirming reversibility under neutral conditions;
- Control experiments ruled out confounds from irrelevant lexical noise;
- Spontaneous stance distributions at isolated nodes disproved the hypothesis that discretization is an inevitable artifact of sequential questioning;
- Strict lexical consistency across cross-model trials eliminated phrasing variation as an explanation for observed jumps.

Collectively, these results demonstrate a systematic, predictable, intervenable, and reproducible dependency between filtering conditions and semantic stances. This dependency satisfies key criteria for causal interpretability in behavioral AI research—not by asserting internal mechanisms, but by establishing reliable input–output–reasoning regularities under controlled perturbation.

5.6.1 Criteria for Behavioral Causality

In behavioral science, establishing causal relationships typically requires satisfying five core criteria: (1) covariation between cause and effect, (2) temporal precedence of the cause, (3) cross-object replicability, (4) portability of the intervention logic, and (5) reversibility of the effect—while simultaneously ruling out plausible alternative explanations. Our experimental design in Section 4 fulfills all these criteria at the behavioral level:

- **Covariation & temporal precedence:** Systematic manipulation of filtering conditions reliably produced directed stance migrations, with changes in input preceding shifts in output.
- **Cross-object replicability:** The same filtering condition (e.g., "nominalism/formalism") induced directionally consistent stance jumps across all three models—Qwen, DeepSeek, and Doubao.
- **Portability:** A questioning protocol distilled from Qwen successfully activated analogous semantic trajectories in DeepSeek, and when re-imported, uncovered latent paths in Qwen itself.
- **Reversibility:** Reversion probes consistently returned all models' discreteness ratings to the fully continuous interval (0.0–0.1), demonstrating that stance shifts are not permanent state changes.

Furthermore, alternative explanations were rigorously excluded:

- Control experiments (Section 4.2) ruled out responses driven by irrelevant lexical noise;
- Spontaneous rating distributions at isolated nodes disproved the hypothesis that discretization is an inevitable artifact of sequential questioning;
- Strict lexical consistency across cross-model trials eliminated phrasing variation as a confound.

Together, these results constitute strong evidence for a *behavioral-level causal relationship* between filtering conditions and semantic stances—one that is predictable, intervenable, and reproducible.

5.6.2 Boundaries and Unruled-Out Alternative Explanations

Despite the robustness of the evidence, several important boundaries must be acknowledged:

- **Black-box limitation:** This study operates entirely at the input–output interface. It does not examine internal activations, parameter dynamics, or computational graph pathways. Consequently, "behavioral causality" here refers strictly to observable dependencies—it makes no claims about the model's internal representational or computational mechanisms.
- **Questioning-order dependence:** All experiments employed a fixed progressive sequence. Although reversion probes demonstrated reversibility, we cannot exclude the possibility that alternative questioning orders might yield different migration trajectories. Future work could address this via randomized or counterbalanced probing sequences.
- **Lexical specificity vs. novelty effects:** It remains unclear whether key terms (e.g., "nominalism," "structuralism") activate stances due to their precise philosophical semantics or simply because they signal conceptual novelty. This study did not include synonym-substitution or semantic-paraphrase controls. Term-variant experiments (e.g., replacing "nominalism" with "anti-realism" or non-philosophical neologisms) would help disentangle semantic specificity from general sensitivity to unfamiliar vocabulary.

5.6.3 Outlook: From Behavioral Causality to Structural Cartography

In summary, this study provides strong evidence for a behavioral-level causal relationship between filtering conditions and model semantic stances—a dependency that is predictable, intervenable, and reproducible. The confirmation of behavioral causality marks not a conclusion, but the inception of a systematic program: the structural cartography of hidden knowledge in LLMs. Specific boundary conditions and concrete future directions are detailed in Section 7.

Establishing behavioral causality is not an endpoint, but a methodological foundation for deeper inquiry. Once a manipulable dependency between filtering conditions and semantic stances has been validated, researchers can treat filtering conditions as calibrated probes to systematically chart the cross-model semantic structure of hidden knowledge in large language models (LLMs). The "semantic coordinates" introduced in this study—such as nominalism/formalism, structuralism, Platonism, and classical calculus background—constitute only a preliminary scaffold of this emerging map.

6 Boundaries and Future Directions

Section 5 evaluated the evidential strength of this work through the lens of behavioral causal inference and underscored the methodological utility of the tracking perspective. This section formally articulates the boundary conditions of our findings and proposes targeted avenues for future research grounded in these limitations.

6.1 Boundary Conditions

Protocol Dependency. All empirical observations are contingent upon the specific measurement apparatus employed: the discreteness rating scale, the sequential questioning protocol, and the operational definitions of filtering conditions. The semantic tiering of the scale, the lexical formulation and ordering of probes, and the selection of "semantic shadows" collectively constitute the *measurement context* of this study. We make no claim that identical results would obtain under alternative scales, phrasings, or protocols. The contribution lies not in proposing a uniquely correct instrument, but in delivering a *reusable, transparent framework* that others can adapt, refine, or challenge.

Limitation of Concept Families. This study focused exclusively on two mathematical concept families: *derivative* (representing continuity-dominant ontologies) and *group* (representing discreteness-dominant ontologies). These were selected to maximize contrast and facilitate the observation of stance variation across conceptual types. However, we do not assert that the observed patterns necessarily generalize to other mathematical domains—such as topology, measure theory, or category theory—nor to non-mathematical concepts. While the *framework* is designed for broad applicability, its empirical scope remains delimited; cross-domain validation is an essential task for future work.

Boundary of Black-Box Observation. All experiments were conducted via end-to-end interaction through public APIs or web interfaces, without access to internal activations, parameters, or computational graphs. Consequently, references to a "high-dimensional semantic space," "hidden knowledge regions," or "sampling paths" are *theoretical constructs* invoked to interpret observed rating distributions—not claims of direct observation. Similarly, metrics such as backward reset distance (BRD) and stance persistence ratio (SPR) reflect operationalized differences within our protocol; they are not measurements of internal representational states.

6.2 Future Work

Future work can advance along two complementary axes:

1. *Horizontal expansion*: Test the universality and transferability of these semantic coordinates across a broader range of mathematical domains (e.g., topology, probability theory, category theory) and diverse model architectures.

2. *Vertical deepening*: Deploy finer-grained questioning ladders, synonym-variant controls, and perturbation-based probing to resolve the internal substructure within a single coordinate (e.g., distinguishing formalist subtypes or mapping gradients between structuralism and Platonism).

Building on the foundation of behavioral causality established in this study, future research can advance along three interrelated directions:
(1) Horizontal Expansion—Cross-Concept and Cross-Model Generalization. Extend the questioning protocol to additional mathematical concept families—such as topology, probability theory, and category theory—and apply it across a diverse set of LLM architectures (e.g., decoder-only, encoder-decoder, multimodal). This will test whether the proposed semantic coordinates (e.g., nominalism, structuralism) exhibit cross-concept stability and whether access-threshold distributions follow consistent topographic patterns across models.
(2) Vertical Deepening—Granular Probing and Lexical Specificity. Develop more refined questioning ladders (e.g., extending Doubao's ten-step progression to finer semantic gradations) and implement systematic term-variant controls—such as synonym substitution, paraphrastic rewording, or replacement with non-philosophical neologisms. These interventions will help disentangle whether stance shifts arise from precise philosophical semantics or from generalized responses to lexical novelty, thereby resolving the internal substructure of individual semantic coordinates.
(3) Cross-Validation via Internal Interpretability. Integrate behavioral probing with internal interpretability techniques—such as activation patching, attention head ablation, or representational similarity analysis—to examine how filtering conditions modulate internal representations within the Filtering–Projection framework. Such work would provide mechanistic cross-validation for behavioral causality. Crucially, this would mark a methodological transition: from *descriptive empirical cartography* (mapping input–output dependencies) to *mechanistic interpretability* (tracing causal pathways in latent space).

7 Conclusion

This study began with a deceptively simple question: *Does a large language model possess a stable "inherent cognition" regarding the discreteness of mathematical concepts?*

Through systematic testing, controlled intervention, reverse probing, and cross-model comparison, we arrived at an answer far richer than anticipated: the model's "cognition" is not a fixed point, but a dynamic, condition-dependent *map*.

Under different filtering conditions, the same model accesses distinct regions of its hidden knowledge terrain—each yielding self-consistent yet divergent stances. Snapshot analyses capture only local projections; only through tracking can we discern the contours of the underlying topography.

Our contribution is thus twofold: first, we demonstrate that semantic stances in LLMs are *manipulable, predictable, and reproducible* under controlled filtering—a form of behavioral causality; second, we propose a shift in explanatory goals: from seeking "the model's view" to mapping "the space of views the model can enact."

In doing so, we do not claim to have opened the black box—but we have begun to chart its shores.

On this map lie distinct topographic features: the dense plains (rating interval 0.0–0.1) that emerge spontaneously under minimal filtering; the rugged peaks (0.7–1.0) revealed only after layered probing

logic strips away surface-level assumptions; the directed stance migrations induced by switching filtering conditions; and the measurable residues of path dependence left in their wake.

The baseline topographies of DeepSeek and Qwen exhibit morphological divergence: DeepSeek's distribution is broad and diffuse, whereas Qwen's is tightly concentrated yet punctuated by high-discreteness outliers. Doubao, serving as an independent validation model, exhibited the highest access threshold for the same semantic coordinate—achieved through its ten-step questioning ladder—thereby providing critical support for the methodological completeness and robustness of our framework.

Collectively, all three models confirm a central insight: hidden knowledge in LLMs is organized around *cross-model shared semantic coordinates*, differing not in the existence of these coordinates, but in the *thresholds required to access them*.

The researcher's task, therefore, is not to locate "the single correct point," but to *chart the map*: under which filtering conditions does the model access which region of its semantic terrain? Moreover, researchers can actively manipulate these conditions to excavate latent semantic paths from low-frequency regions—and even transplant them across models. This capability constitutes the core innovation of our *Filtering–Projection* approach.

Using a reusable, protocol-driven methodology, we systematically manipulated filtering conditions and recorded—and successfully transplanted—the semantic stances of three distinct models on the concepts of *derivative* and *group*. The observed covariation between intervention and response, cross-object replicability, cross-model portability, and reversibility upon reversion probes jointly establish a dependency that is systematic, predictable, and intervenable.

We make no claim to have uncovered the model's "true cognition," nor to have exposed its internal computational mechanism. The model's response is always only the part of the answer you are able to ask out. As for the rest—they have always been there, waiting to be asked.

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